

The empirical recurrence for OEIS A251347: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A251347 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 243$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A251347 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with every 2×2 subblock summing to 3 or 4 or 5.”. It has offset 1 and begins

11709, 663307, 39019417, 2324761359, 139048718681, 8328173688589, 499036482452523, 29907963638879071, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 10:22:57, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 243$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 486$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	114	c_{16}	−41326282472832336	c_{31}	−4251561494071989583663	c_{46}	−16517705303
c_2	−3736	c_{17}	59991498383146496	c_{32}	2890101987419181989691	c_{47}	80877928241
c_3	10309	c_{18}	468775436294232225	c_{33}	4976886982569376978578	c_{48}	10234795169
c_4	1541974	c_{19}	−1045134580270539760	c_{34}	−4497110141761126817828	c_{49}	−8687631134
c_5	−21668945	c_{20}	−3546793895262289288	c_{35}	−3983022366412924843977	c_{50}	317278555
c_6	−130798525	c_{21}	10638815228094880546	c_{36}	4722085541717364879158	c_{51}	5204867189
c_7	4250013885	c_{22}	17663505897946792061	c_{37}	2021791782496345098891	c_{52}	−348831847
c_8	−10584217483	c_{23}	−71532864737224651816	c_{38}	−3425203116416679118808	c_{53}	−167639137
c_9	−290099476132	c_{24}	−53607662369679072293	c_{39}	−487888951588705496190	c_{54}	172211157
c_{10}	1676479096177	c_{25}	332687747659453821743	c_{40}	1714295030945509788328	c_{55}	267104629
c_{11}	9070017412782	c_{26}	64948254189598136430	c_{41}	−91974574482056409350	c_{56}	−3212478
c_{12}	−85698058199107	c_{27}	−1092737305153041046413	c_{42}	−580222921936235036069	c_{57}	−1695989
c_{13}	−113454531436992	c_{28}	193704615084993432657	c_{43}	117187236925115869920	c_{58}	2121965
c_{14}	2387253217731570	c_{29}	2555529214411365139674	c_{44}	127109717596886929588	c_{59}	224952
c_{15}	−1007925751667991	c_{30}	−1155151329915707349024	c_{45}	−42286604106995109767	c_{60}	−30670

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251347.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.